RETAIL SALES ANALYSIS AND BUSINESS INSIGHTS

Thiranex Internship — Project 4

1. Introduction

Retail sales analysis helps businesses understand customer purchasing patterns, product demand, and revenue performance. This project analyzes online retail transaction data using Python to identify sales trends and generate insights that can support business decisions.

2. Project Objectives

- Analyze retail transaction data.
- Identify and handle data-quality issues.
- Calculate sales revenue.
- Identify the best-selling products.
- Analyze monthly sales trends.
- Compare revenue across countries.
- Create visualizations and provide business recommendations.

3. Dataset Description

The project uses the Online Retail dataset obtained from Kaggle.

Dataset: Online Retail

Source: https://www.kaggle.com/datasets/vijayuv/onlineretail

The dataset contains retail transaction information, including invoice numbers, product descriptions, quantities, invoice dates, unit prices, customer identifiers, and countries.

4. Tools and Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

5. Methodology

5.1 Data Loading

The dataset was loaded into Jupyter Notebook using Pandas.

5.2 Data Exploration

The dataset was examined to understand its dimensions, column names, data types, missing values, duplicate records, and summary statistics.

5.3 Data Cleaning

Duplicate records were removed, rows with missing product descriptions were excluded, and records with non-positive quantities or unit prices were filtered out for positive-sales analysis.

5.4 Revenue Calculation

Revenue was calculated for each transaction line using:

Revenue = Quantity × Unit Price

5.5 Exploratory Data Analysis

The following analyses were performed:

- Best-selling products by quantity.
- Monthly revenue trends.
- Revenue by country.
- Distribution of transaction revenue.

6. Visualizations

Four visualizations were created to support the analysis:

1. **Top 10 Best-Selling Products:** Compares products by total quantity sold.
2. **Monthly Sales Trend:** Shows how revenue changes over time.
3. **Top 10 Countries by Revenue:** Identifies countries contributing the most revenue.
4. **Revenue Distribution:** Shows the distribution of transaction-line revenue.

7. Key Findings

The notebook results should be used to complete the following findings:

- **Best-selling product:** Insert the product with the highest total quantity.
- **Highest-revenue country:** Insert the country with the highest revenue.
- **Best-performing month:** Insert the month with the highest revenue.
- **Total analyzed revenue:** Insert the calculated amount and explain which transactions were included.
- **Data quality:** Summarize the missing values and duplicates identified.

Note: Actual numerical findings must be copied from the notebook output. They should not be estimated or invented.

8. Business Recommendations

Based on the verified analysis, the business could consider the following actions:

- Maintain adequate inventory for products with high demand.
- Plan stock levels ahead of seasonal sales peaks.
- Investigate countries with strong revenue contributions for potential growth opportunities.
- Monitor unusual transaction values.
- Review cancelled invoices and returned products to improve sales reporting.

These recommendations should be refined to match the actual findings.

9. Limitations

The analysis filters out non-positive quantities and prices for positive-sales analysis. However, cancelled invoices and returned goods may affect revenue calculations. The resulting revenue total should therefore be treated as a preliminary transaction-line measure rather than final net revenue.

Some customer identifiers may be missing even when the transaction itself is valid.

10. Conclusion

This project demonstrates how Python can be used to analyze retail transaction data, clean records, calculate revenue, identify product demand, and visualize sales trends.

The analysis provides a foundation for understanding retail performance and making data-informed business decisions. Verifying the numerical findings and reviewing cancellations and returns are important final steps before submitting the project.

11. Project Deliverables

- Jupyter Notebook containing the analysis.
- Cleaned retail dataset.
- Top 10 best-selling products chart.
- Monthly sales trend chart.
- Top 10 countries by revenue chart.
- Revenue distribution chart.
- Project report.
- README file for GitHub.